

State-Level Analysis of Per-Capita Medicare Spending on PwD & Pw/oD

**Guillermo Castaneda & Aayushi
Garg**



Introductions



Guillermo Castaneda

- **B.S. Public Policy & Data Analytics '27 & MPA '27**
- **Los Angeles native (Boyle Heights & South Central)**
- **Strategy, Development & Research**
- **Basketball is basically my second major**



Aayushi Garg

- **B.S. Public Policy & Data Analytics '28 & MPP '28**
 - **Los Angeles native**
- **Fascinated with intersection of technology & policy**
 - **Huge music nerd**

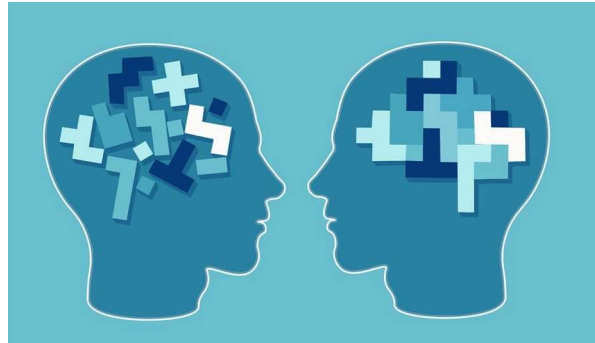
Research Question

How does per-capita Medicare spending differ across states for individuals with dementia compared to those without dementia, and how have these spending trajectories changed between 2011 and 2021?

Background and Motivation

Why does this matter?

- 5.6 million people live with dementia in the US, with 5.0 million being 65+ (US Cost of Dementia, 2025).
- Dementia is among the costliest conditions in the U.S. Medicare spends roughly \$10,000 more per person annually on dementia patients than on comparable non-dementia patients (Alzheimer's Association, 2024).
- As the population ages, these costs will escalate. Texas's 65+ population alone is projected to more than double by 2050 (Texas 2036, n.d.)
- States differ dramatically in demographics, Medicaid policy, and care infrastructure:



Data and Methods

Data

- Source: Medicare fee-for-service claims data
- Benchmark years: 2011, 2016, 2021
- Populations: Patients with dementia (PwD) and patients without dementia (Pw/oD), controlled for age, sex, race, and comorbid conditions making the two groups directly comparable
- Cost components: Inpatient (IP), Skilled Nursing (SNF), Home Health (HHA), Hospice (HOS), Outpatient (OP), Durable Medical Equipment (DME), Prescription Drugs (Part D)

State Selection

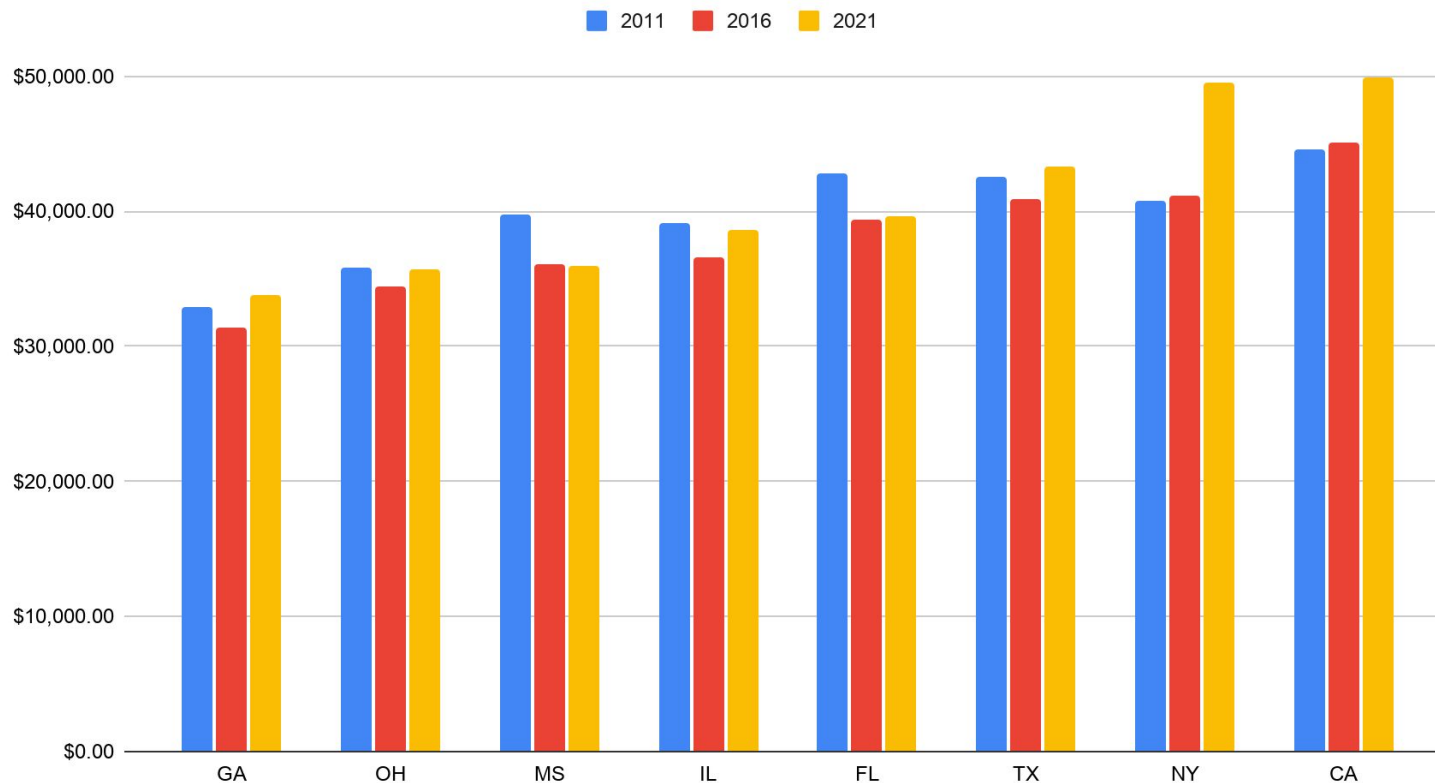
- Eight states chosen for demographic and policy diversity:
 - High-spending, policy-progressive: California, New York, Illinois
 - Large/rapidly aging populations: Florida, Texas
 - Midwestern baseline (Medicaid expansion): Ohio
 - Southern, rural, resource-constrained: Mississippi, Georgia

Methods

- Calculated mean per-capita spending by state, year, and dementia status using Stata
 - Generated descriptive statistics (means, standard deviations, min/max) by service category
 - Compared spending distributions across states and at different points in time via tables and visualizations

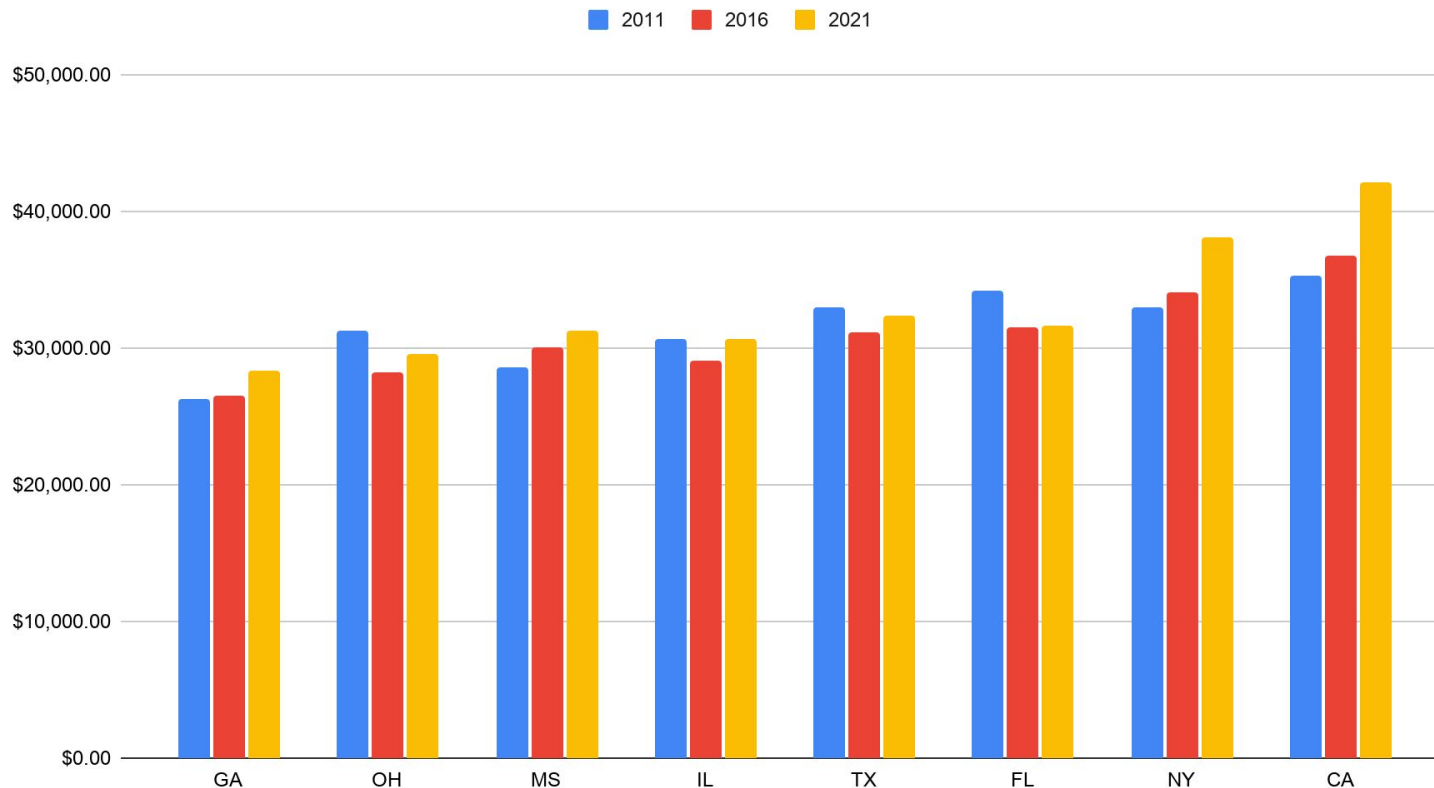
Results: State-Level Cost Trends (Dementia)

Average Per Capita Medicare Cost by State for PwD (2011 - 2021)



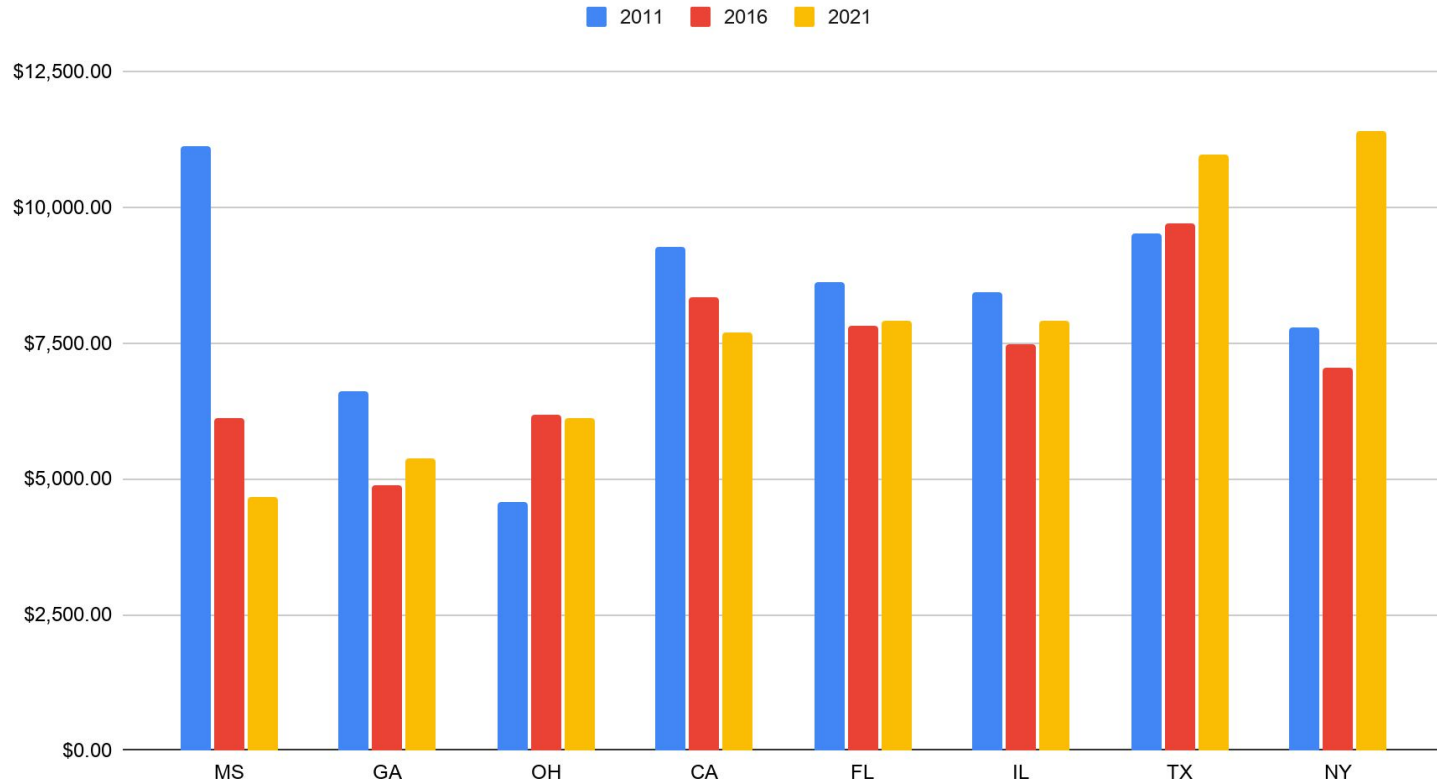
Results: State-Level Cost Trends (Non-Dementia)

Average Per Capita Medicare Cost by State for Pw/oD (2011 - 2021)



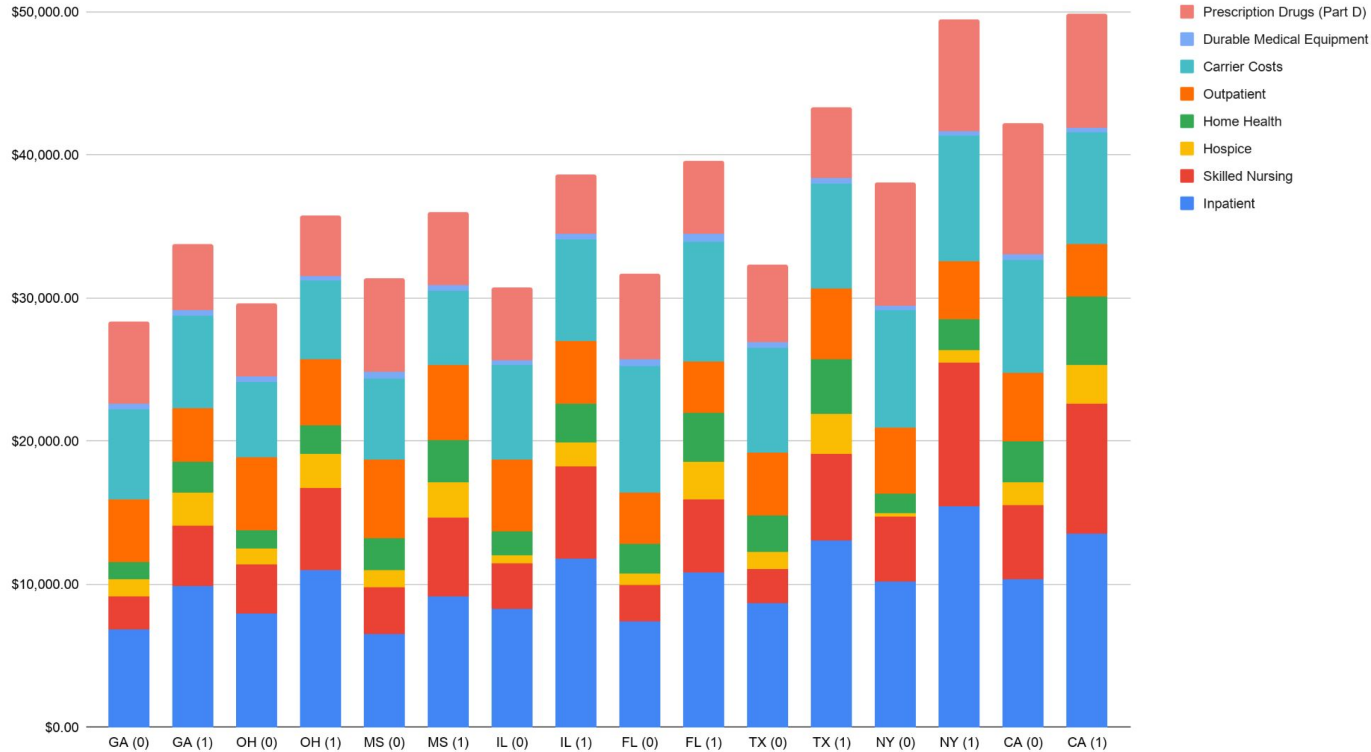
Results: State-Level Cost Trends (Dementia vs Non-Dementia)

Net Medicare Cost Difference for PwD vs. Pw/oD (2011 - 2021)



Results: Cost Category Breakdown

Medicare Cost Category Breakdown (2021)



Discussion

PwD vs. Pw/oD Cost Differences

- PwD cost ~\$10,000/year more than comparable Pw/oD driven primarily by skilled nursing facility and home health utilization
- National spending shift: inpatient and SNF costs declining; outpatient and Part D costs rising affecting PwD disproportionately given their higher care intensity
- Persistent state-level gaps: CA and NY spend 15-35% more on average, with some states approaching 50% differences

Discussion

Why PwD Costs Differ by State

Demographics

- States with older (FL: ~21% aged 65+) and faster-growing elderly populations (TX, GA, OH) face larger PwD cohorts and cost exposure

Geography & Access

- Regional wage adjustments create inherent price differences; rural states like MS have fewer specialists, constraining high-cost services for PwD

Policy Environment

- CA and NY expanded home-based dementia care; OH expanded Medicaid (2014); MS has not, potentially limiting access for low-income PwD

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